

MicroNotes Project Report

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Assignment: Build MicroNotes App

1. What is the Difference Between Frontend and Backend?

The frontend is the client-side of the application that runs directly in the user's browser, responsible for rendering the User Interface (UI), capturing user actions, and managing local component state (built using HTML, CSS, and React in this app). In contrast, the backend is the server-side that runs on a remote host (built with Node.js and Express here), handling business logic, authentication, and hosting the data store. While the frontend presents information visually and provides a user-friendly layout, the backend serves as the single source of truth that processes API requests, performs calculations, and decides what data to send back. They communicate over the network using standard protocols like HTTP via structured REST endpoints.

2. What Does async/await Actually Do?

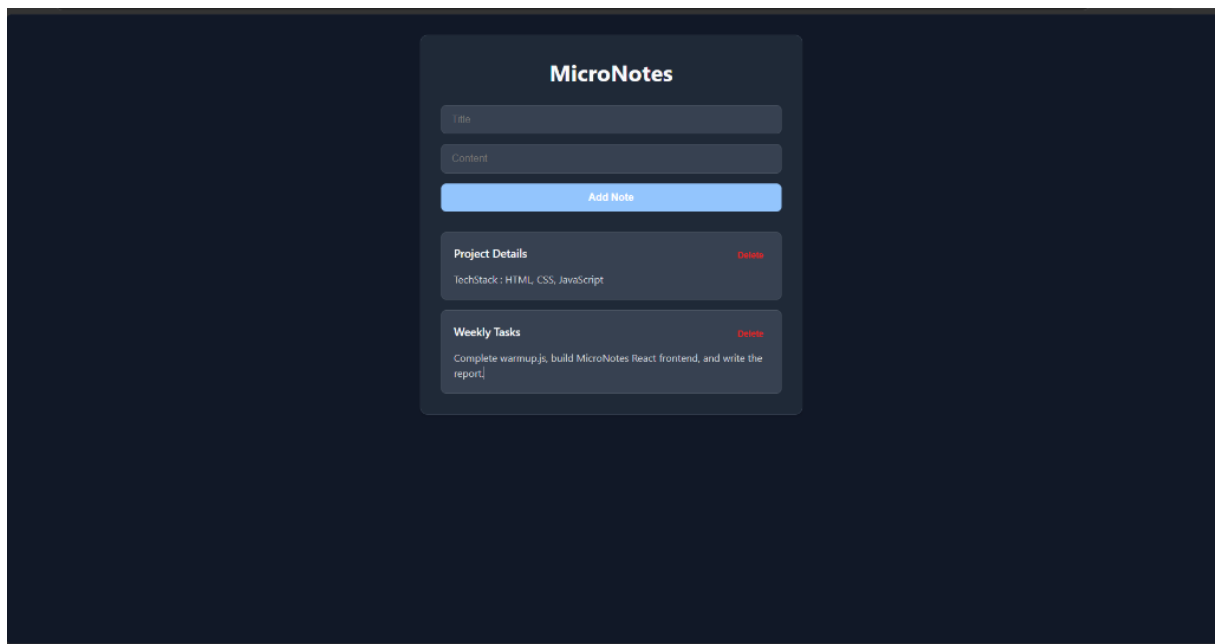
In JavaScript, asynchronous code is used to perform operations that take time (such as fetching data from a server) without freezing the entire browser interface. A Promise represents a value that will become available in the future.

The `async/await` syntax is a clean, readable way to work with these Promises. By marking a function as `async`, we are allowed to use the `await` keyword inside it. The `await` keyword pauses the execution of that specific function until the Promise resolves, returning the actual result directly. This avoids nesting multiple callback blocks (often referred to as 'callback hell') and keeps the codebase highly maintainable.

Example implementation from App.jsx:

```
const fetchNotes = async () => {
  try {
    setLoading(true);
    // 1. Await pauses here until the network request completes and returns the response
    const response = await fetch("http://localhost:5000/api/notes");
    // 2. Await pauses here until the response body is parsed into JSON
    const data = await response.json();
    setNotes(data);
  } catch (error) {
    console.error("Error fetching notes:", error);
  } finally {
    setLoading(false);
  }
};
```

3. App Screenshot (Showing 2 Notes)



4. One Thing That Was Confusing and How I Figured It Out

A particularly confusing issue occurred when trying to push the project commits to the remote GitHub repository. Git rejected the push command with a 403 Forbidden error stating that permission was denied to a previously cached account ('va7tech-coder').

How it was resolved:

1. Checked Git settings using 'git config --local --get-regexp user' and confirmed the local repository details were correct, but the system credentials manager was still injecting the wrong account token.
2. Generated a Personal Access Token (PAT) with full 'repo' access scopes from the 'raybuilds' GitHub account.
3. Updated the Git remote origin URL to include the token: 'git remote set-url origin https://<PAT_TOKEN>@github.com/raybuilds/micronotes.git'.
4. Executed the push again, which successfully verified writing credentials. This highlighted the distinction between Git author commit metadata and network push authorization credentials.